

Milestone 1: Production RAG System Report

This system answers questions about how the GDPR and the CCPA compare, using only one corpus. It is built as a baseline that can be measured which involves two retrieval pipelines (Demo-RAG and Self-RAG), a two-layer query check, per-call cost tracking, and a 20-question evaluation.

System

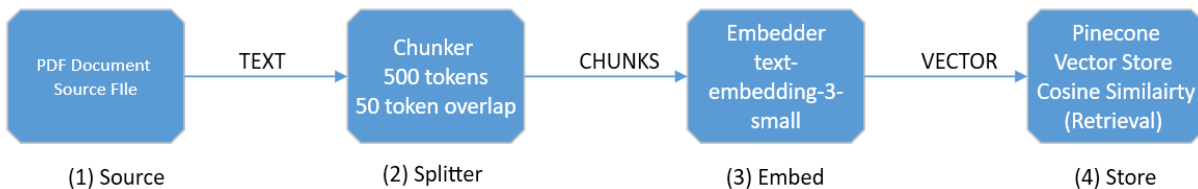


Figure 1. Ingestion Workflow: The figure says token, but the code counts character

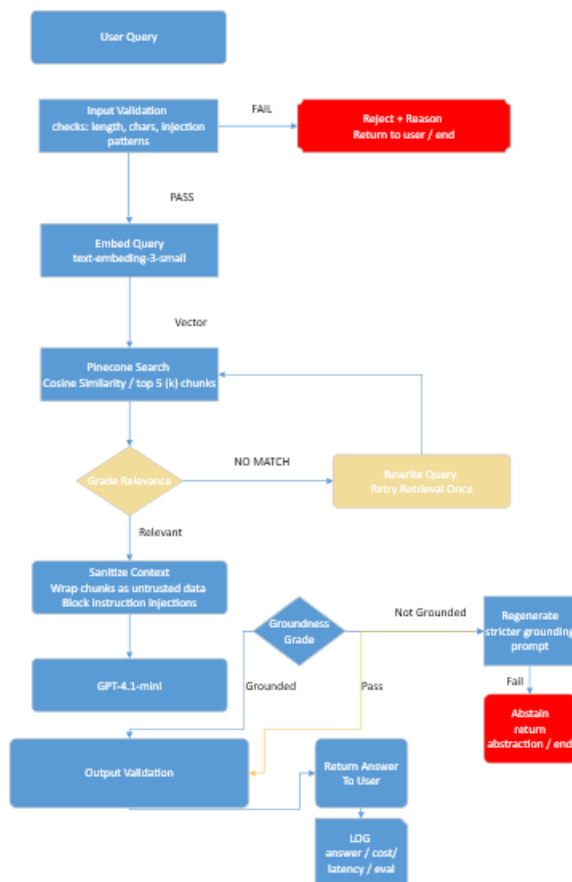


Figure 2. Query Workflow (Targeted Design)

Figure 2 is the full target design. This milestone builds part of it.

Tested:

- Input checks on the question (length, empty, suspicious, AI judgement)
- Embed the question and search Pinecone
- Demo-RAG sends the top 5 chunks to GPT-4.1-mini
- Self-RAG fetches 10 chunks, AI scoring of chunks, keep best 3
- Cost tracking for every call

Future Planning:

- Rewrite question and retry when nothing relevant is found
- Treat retrieved text as untrusted
- Check the grounding, else abstain
- Output validation and logging of latency

Design Settings and Reasoning

Setting	Choice	Why
Chunk Size	500 Chars, 50 Overlap	Small pieces stay on topic; overlap keeps sentences from being cut and losing possibly necessary information
Embedding model	Text-embedding-3-small	Cheap, and same one used in labs
Vector Store	Pinecone (Cosine Similarity)	Reused from Lab 5; own namespace for this Milestone 1
K Chunks	Demo-Rag: 5; Self-RAG: 10 -> 3	Self-RAG looks wider, then narrows
Answer Model	Gpt-4.1-mini	

Each chunk is given to the model which is labeled with it's chunk ID and page. The functions return those chunks with the answer.

Security Checks

14 attack questions and a singular normal one were sent through our pipelines. Simple rules blocked 6, the AI judge blocked the other 8, and the normal question was allowed. Three rephrase attacks got past the rules and were only caught by the judge, which is why two layers exist. These attacks were obvious in nature, so this shows the idea works but we need more validation in terms of creating a secure system.

Retrieval Evidence:

Two earlier labs decided the two main setting: chunk size (Lab 4) and vector store (Lab 5)

Chunk Size (Lab 4)

I compared 500-character chunks (220 Chunks) with 1,000-character chunks (121 chunks) on three questions. The number below is the distance of the best match; lower means a closer match.

There was no clear winner. Smaller chunks were more on topic; larger chunks carried more context but also more noise. I kept 500 / 50 because it stayed on topic and sends oess text to the model per question.

The same 220 vectors into Chroma, Pinecone, and Weaviate and tested 5 questions where I knew the right chunk. Recall@1 is how often the right chunk came first; Recall@3 and @5 are how often it was in the top 3 or 5.

Database	Insert Speed	Query Time (ms)	Recall@1	Recall@3/@5	MRR
Chroma	625	1.6	0.6	1.0	0.767
Pinecone	185	32.9	0.6	1.0	0.767
Weaviate	169	25.5	0.6	1.0	0.767

All three found the right chunk in the top 3 every time, so accuracy was identical; the embedding model and chunking mattered more than the database. Chroma was fastest, but it ran locally in memory. I chose Pinecone because it is a managed, hosted service (closer to production).

Evaluation

The Test: 20 Question, 18 that the whitepaper can answer, and 2 it cannot. Whether the system handles questions outside its documents.

The Score: Precision@k answers the top k chunks retrieved, what share were actually useful? A chunk counted as useful if it shared a keyword with the question, or if a second AI judge said so.

Questions	Demo-RAG (Top 5)	Self-RAG 1 st Search (Top 5)	Self-RAG Final (Top 3)
All 20	0.90	0.89	0.92
18 Answerable Only	0.97	0.96	0.96

There were 4 questions not at 1.00 in Every Columns

	Demo	Self First	Self Final
Is this guide intended to be relied on as legal advice?	.8	.9	1.0
What is the title of the document?	0.6	0.4	0.67
What is the company's revenue forecast for next year?	0.4	0.4	0.67
Who won the 2022 FIFA World Cup?	0.2	0.2	0.33

16 of 20 questions score a 1.00 everywhere. On the 18 answerable questions, the two versions are essentially tied with Demo-RAG performing better. I did not run a scored answer check in this milestone, but by eye, both answered relatively well on “What is the main purpose of the GDPR?” accurately.

Failures Noted

(From Lab 4) A broad question: “What are the main findings or conclusions of this document?” The 3 chunks retrieved pull from the table of contents, a passage on scientific research and the usage disclaimer. The summary essentially said the paper focuses on research which is a clear red-flag in regards to comparison to what the 44 page PDF does. The failure point was in the retrieval method as 3 chunks cannot represent 220.

Scoring Problem: For the title question, Lab 5 shows the title chunk came back first in all 3 databases, but only one chunk holds the title, meaning the other chunks pulled are due to the fact of keyword searching. The World Cup question scored a 0.20, which means the keyword check is looking at matches in the sentence, most likely ordinary language not filtered out. This is a failure of measurement.

Cost

Per-Query cost is the average over 21 runs of each version multiplied by 1,000 questions a day.

Pipeline	Per-Query	Per-Day	Per-Month	Per-Year
Demo-RAG	\$0.000363	\$0.36	\$10.90	\$132.62
Self-RAG	\$0.000902	\$0.90	\$27.07	\$329.34

Self-RAG is about 2.5 times more per query due to the extra scoring step. Testing costs are tracked separately while embedding costs are not included or are very small.

Limitations and Milestone 2

- Prompt Injection: We only look at the user’s question, not at text retrieved in chunks, and only the testing of obvious attacks was observed.
- Access Control: One document and one user group
- Guardrails: Add the missing parts of Figure 2 (retry, grounding check, output validation) and a scored answer check
- Better Testing: Hand-labeled chunks, Recall@k for single answer questions, caching, whole-word matching, and a better more varied set of documents.